

Introduction

- • Airbnb is an online marketplace for short-term lodging.
- • This project explores the 2019 New York City Airbnb dataset from Kaggle.
- • Goal: To understand pricing trends, room types, popular locations, and host behavior using EDA techniques.

Dataset Overview

- • Source: Kaggle - Airbnb NYC 2019 Dataset
- • Rows: ~48,000, Columns: 16
- • Contains information about listings, hosts, prices, reviews, and availability.

Main Columns in Dataset

- neighbourhood_group: Borough of NYC (Manhattan, Brooklyn, etc.)
- neighbourhood: Specific area inside borough
- room_type: Type of accommodation
- price: Price per night (USD)
- number_of_reviews, reviews_per_month: Guest activity
- availability_365: Days available for booking
- host_id, host_name: Host information

Data Loading & Inspection

- Code used:
- `df = pd.read_csv('AB_NYC_2019.csv')`
- `print(df.shape)`
- `df.head()`
- Purpose: Load dataset, check shape, and preview first few records.

Data Cleaning

- • Checked for missing values and duplicates.
- • Filled missing 'reviews_per_month' with 0.
- • Removed price ≤ 0 .
- • Ensured data types are correct.
- Outcome: Clean and ready dataset for analysis.

Q1: Which Borough Has the Most Listings?

- • Used `value_counts()` on `'neighbourhood_group'`.
- • Manhattan and Brooklyn have the highest number of listings.
- • Staten Island has the least.
- → Manhattan is the most popular area for Airbnb in NYC.

Q2: Which Borough Has the Highest Average Price?

- • Grouped data by 'neighbourhood_group' and calculated mean price.
- • Manhattan has the highest average price.
- • The Bronx and Staten Island are the cheapest.

Q3 & Q4: Room Type and Price Distribution

- • Majority listings are 'Entire home/apt' and 'Private room'.
- • Price distribution is right-skewed (most affordable, few expensive).
- • Positive skew confirmed using `df['price'].skew()`.

Q5: Price Variation Across Boroughs

- • Used boxplots to compare prices across boroughs.
- • Manhattan shows highest median and outlier prices.
- • Clear price variation between areas.

Q6: Which Neighbourhood Has Most Listings?

- • Williamsburg and Bedford-Stuyvesant (Brooklyn) are top areas.
- • Harlem and Upper West Side popular in Manhattan.

Q7: Relation Between Reviews and Price

- • Used scatterplot between 'price' and 'number_of_reviews'.
- • No strong relationship — cheaper listings tend to have more reviews.

Q8-Q10: Availability, Host Count & Correlations

- • Listings with 0–100 availability dominate (short-term focus).
- • Most hosts have only 1 listing; few have many (superhosts).
- • Correlation heatmap shows price weakly related to availability and reviews.

Key Insights & Conclusions

- • Manhattan dominates Airbnb listings and prices.
- • Room type affects price: Entire homes cost more.
- • Most listings have low availability — short-term stays.
- • Data is right-skewed; log transform helps visualization.
- • Few hosts own multiple properties (commercial use).

Summary & Future Work

- Dataset explored successfully using EDA techniques.
- Future scope: Predict pricing, host analysis, and sentiment analysis from reviews.
- Thank you!